

I am a PhD student at the University of Sydney researching feature-based time series analysis and machine learning in complex systems, with a strong interest in preserving interpretability in data-driven methods. My current work explores how multivariate time series can be represented as graphs, enabling feature-feature relationships to encode domain-invariant structure and information to support flexible representation learning across datasets that differ in size and length, with applications to consciousness in marmosets. More broadly, I am fascinated by the intersection of neuroscience and machine learning and the idea of fundamental computational principles of information, learning, and intelligence. I am also increasingly compelled by how geometric structure in data and models shapes learning.

I completed a Research Associate position following my 2024 Honour's thesis on tensor network optimisation for encoding two-dimensional data into quantum amplitudes. While the exponential dimension of Hilbert space offers extraordinary computational potential, it also represents a curse of dimensionality for quantum state preparation. This work explored how one-dimensional tensor networks can approximate image data and discretised functions in highly compact forms, deepening my interest in how structure in complex data enables compressed representations.

I am exceptionally motivated to attend EEML 2026 to further my understanding of deep and reinforcement learning. The experimental sessions would broaden my skillset as a machine learning researcher. I am especially excited by the opportunity to learn from new work at Isomorphic Labs through Andreea Deac, as well as applications of machine learning in medicine and neuroscience. I would also value the opportunity to connect with leading researchers, particularly Federico Barbero and Dr Petar Veličković, whose expertise in geometric deep learning closely aligns with my interests, and with early-career students and researchers.